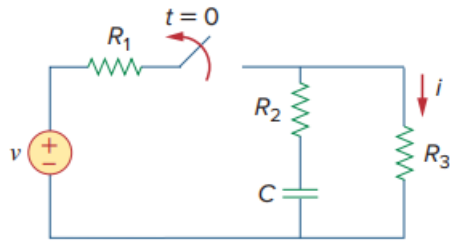


Problem

Using Fig. 7.85, design a problem to help other students better understand source-free RC circuits.

Fig. 7.85:



Step-by-step solution

Step 1 of 6

Refer to figure 7.85 given in the textbook.

Assume that the voltage source is of 24 V, and the capacitance value as $\frac{1}{3}$ F; and the value of resistances R_1 , R_2 , and R_3 are $2\ \Omega$, $5\ \Omega$, and $4\ \Omega$ respectively.

Draw the circuit for $t = 0^-$ as shown in figure 1.

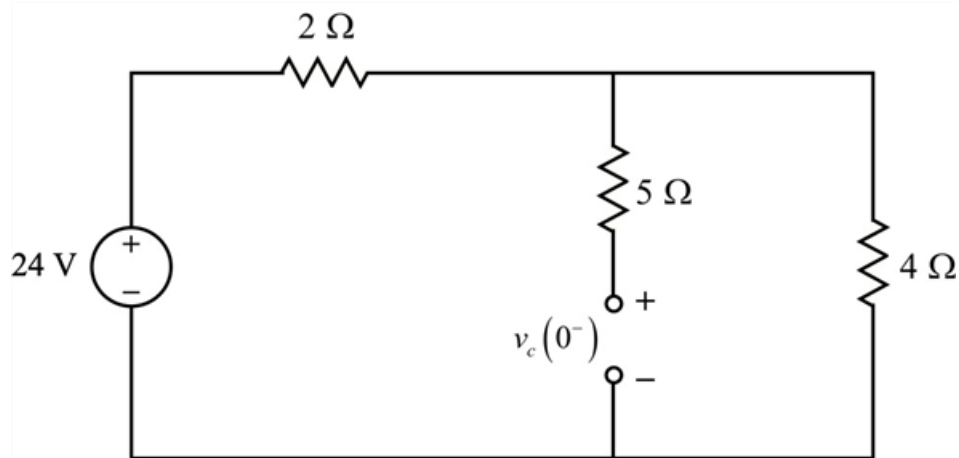


Figure 1

Step 2 of 6

Determine the initial voltage $v_c(0^-)$ across the capacitor.

Apply voltage division rule.

$$\begin{aligned} v_c(0^-) &= \frac{4}{4+2} \times 24 \\ &= \frac{4}{6} \times 24 \\ &= 16\text{ V} \end{aligned}$$

Step 3 of 6

Capacitor doesn't allow sudden changes in the voltage.

$$v_c(0^-) = v_c(0^+) = v_c(0) = 16 \text{ V}$$

Here, $v_c(0)$ is the voltage across the capacitor after the switch is being closed.

Step 4 of 6

Redraw the circuit for $t = 0^-$ as shown in figure 2.

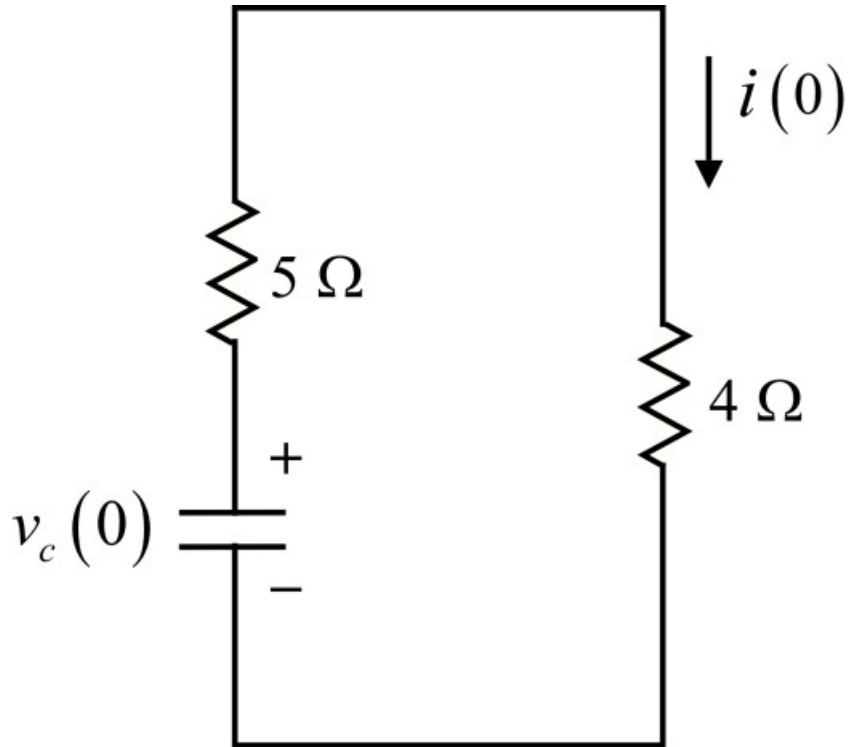


Figure 2

Step 5 of 6

Apply Kirchhoff's voltage law to the circuit shown in figure 2.

$$v_c(0) - 5i(0) - 4i(0) = 0$$

Substitute 16 for $v_c(0)$.

$$16 - 5i(0) - 4i(0) = 0$$

$$9i(0) = 16$$

$$i(0) = \frac{16}{9}$$

$$i(0) = 1.77 \text{ A}$$

Step 6 of 6

Calculate the time constant τ .

$$\tau = RC$$

Substitute 9 for R , and $\frac{1}{3}$ for C .

$$\begin{aligned}\tau &= 9 \times \frac{1}{3} \\ &= 3 \text{ s}\end{aligned}$$

The expression for the current $i(t)$ is,

$$i(t) = i(0)e^{\frac{-t}{\tau}}$$

Substitute 1.77 for $i(0)$, and 3 for τ .

$$i(t) = 1.77e^{\frac{-t}{3}} \text{ A}$$

Therefore, the current $i(t)$ is $\boxed{1.77e^{\frac{-t}{3}} \text{ A}}$.